# Scale-Dependent Information Units as Phase Transitions on $T^4$

## Geometry as the Primary Foundation of Information and Energy

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# Abstract

*Dok. 255 derived Landauer's principle and Vopson's mass-information equivalence from $T^4$ geometry, identifying information with topologically frozen kinetic energy. The present refinement shows that this derivation holds as a special case at the thermodynamic scale, not universally. The more fundamental picture is: the minimal information unit is by definition the topological winding quantum $\Delta w = 1$ — scale-universal and energy-independent. The corresponding energy $E_{bit}(L) = \hbar c/L$ follows from action and the uncertainty relation and is scale-specific. Landauer's $E_L = k_B T \ln 2$ is the special case at the temperature-length scale where $E_{bit}(L) = k_B T \ln 2$ — biological room temperature. All phase transitions are discrete because $\pi_1(T^4) \cong \mathbb{Z}^4$ admits no half-integer winding quanta. Geometry is more fundamental than energy. We work in natural units with $c = \hbar = k_B = 1$ unless stated otherwise.*

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## .1 The Problem: Three Levels, One Hierarchy

Dok. 255 stated:

*Information is topologically frozen kinetic energy.*

This is correct as a description of the thermodynamic regime. The deeper question is: what is more fundamental — geometry or energy?

The internal consistency check shows that the equivalence temperature at which $E_{\text{bit}}(L) = k_B T \ln 2$ holds varies over **70 orders of magnitude**:

| **Scale** | **L [m]** | $T_{\text{equiv}}$ **[K]** |
|---|---|---|
| Planck | $1.6 \times 10^{-35}$ | $2.0 \times 10^{32}$ |
| FFGFT $L_0$ | $2.1 \times 10^{-39}$ | $1.5 \times 10^{36}$ |
| QCD/Proton | $1.0 \times 10^{-15}$ | $3.3 \times 10^{12}$ |
| Atom | $1.0 \times 10^{-10}$ | $3.3 \times 10^{7}$ |
| DNA codon | $1.0 \times 10^{-9}$ | $3.3 \times 10^{6}$ |
| **Cell** | $\mathbf{1.0 \times 10^{-5}}$ | **331** |
| Cosmic | $4.4 \times 10^{26}$ | $7.5 \times 10^{-30}$ |

with the formula (cf. Section .2):

$$T_{\text{equiv}}(L) = \frac{\hbar c}{k_B \, L \, \ln 2}. \tag{.1}$$

Only at the cellular scale does $T_{\text{equiv}} \approx 331$ K match biological room temperature.

## .2 Three Independent Levels

The calculation enforces a separation into three levels:

**Level 1: Topological (universal).**

**Definition .2.1** (Minimal information unit)**.** The minimal information unit on $T^4$ is the winding quantum $I_{\text{bit}} = \Delta w = 1$, $\Delta w \in \mathbb{Z}$. One bit is identified with a minimal topological winding difference $\Delta w = 1$.

This identification is the conceptual bridge between topology and information theory. The unit is scale-independent: $\pi_1(T^4) \cong \mathbb{Z}^4$ enforces discrete integer winding numbers at every scale. No phase transition can produce half a winding quantum.

**Level 2: Geometric (scale-specific).** The energy of a winding quantum at length scale $L$ follows from action and the uncertainty relation: The action of one winding quantum is $S = \Delta w \cdot h = h$. From the uncertainty relation $\Delta x \cdot \Delta p \sim \hbar$ at spatial scale $L$ one obtains $p \sim \hbar/L$, hence:

$$E_{\text{bit}}(L) = pc = \frac{\hbar c}{L}. \tag{.2}$$

*Note:* The numerical prefactor is a scale definition; the formula gives the correct order of magnitude. In natural units ($c = \hbar = k_B = 1$), $E_{\text{bit}}(L) = 1/L$.

**Level 3: Thermodynamic (one scale).** Landauer's principle [2]:

$$E_L = k_B T \ln 2 \quad \text{per bit} \tag{.3}$$

holds in the thermodynamic regime, i.e. when $E_{\text{bit}}(L) = E_L$. Setting equal yields equation (.1).

### Visualisation of the Hierarchy

| Level | Character | Quantity | Validity |
|---|---|---|---|
| 1 | topological | $\Delta w = 1$ | universal, all scales |
| | ↓ | | |
| 2 | geometric | $E_{\text{bit}}(L) = \hbar c/L$ | every scale $L$ |
| | ↓ | | |
| 3 | thermodynamic | $E_L = k_B T \ln 2$ | only when $E_{\text{bit}} = k_B T$ |

## .3 Geometry is More Fundamental than Energy

The derivation hierarchy is:

$$\begin{aligned} T^4\text{-geometry} &\longrightarrow \Delta w = 1 \longrightarrow E_{\text{bit}}(L) = \hbar c/L \\ &\longrightarrow E_L = k_B T \ln 2 \text{ (special case).} \end{aligned} \tag{.4}$$

Energy is a scale-dependent projection of $T^4$ geometry, not its foundation.

| | |
|---|---|
| **Dok. 255** | Information = frozen kinetic energy |
| **Dok. 257** | Information = winding quantum $\Delta w$; energy is a scale-specific projection of it |

*Both statements consistent: Dok. 255 describes the thermodynamic special case.*

## .4 Phase Transitions are Discrete at Every Scale

All phase transitions in physics are discrete. This follows directly from $\pi_1(T^4) \cong \mathbb{Z}^4$.

| Scale | Phase transition | Information unit |
|---|---|---|
| Planck/FFGFT | Topological enclosure | $\Delta w = 1$, $E \sim E_P$ |
| QCD | Confinement, hadronisation | Baryon $B = 1$, $E \sim \Lambda_{\text{QCD}}$ |
| Atom | Electron configuration | Quantum number $n$, $E \sim$ eV |
| Molecule/DNA | Codon, folding | 3 base pairs, $E \sim k_B T$ |
| Cell | Protein folding | Domain, $E \sim k_B T$ |
| Cosmic | $\sigma$-shift | $\hbar c / R_U$, $E \ll k_B T_{\text{CMB}}$ |

At every scale: $I_{\text{bit}} = \Delta w = 1$ (topological). The energy of the bit varies with scale.

## .5 Scale Resonance at the Scale of Life — Hypothesis

For $L \sim 10^{-5}$ m (cell size) and $T \sim 300$ K:

$$E_{\text{bit}}(L_{\text{cell}}) = \frac{\hbar c}{10^{-5}\,\text{m}} \approx 3.2 \times 10^{-21}\,\text{J} \approx k_B \cdot 331\,\text{K} \cdot \ln 2. \qquad (.5)$$

This suggests the following hypothesis:

*Life as a self-organising, information-processing structure may only exist at scales where $E_{\text{bit}}(L) \approx k_B T$. Biological room temperature would then be a consequence of this scale resonance.*

Whether this coincidence is causal or accidental remains open and deserves further investigation. It is stated here as a hypothesis, not a derivation.

## .6 Refinement of Dok. 255

1. **Correct in Dok. 255:** Landauer and Vopson follow from $T^4$ geometry in the thermodynamic regime.

2. **Refinement:** The universal information unit is $\Delta w = 1$ (topological), not $k_B T \ln 2$ (thermodynamic).
3. **Extension:** Energy is a scale-specific projection of geometry.
4. **Consequence:** $I_{\text{bit}} = \Delta w$ holds at all scales. $E_L = k_B T \ln 2$ holds only at the scale of life. $E_{\text{bit}} = \hbar c/L$ holds geometrically at every scale.

# .7 Summary

1. The minimal information unit is by definition the winding quantum $\Delta w = 1$ — universal from $\pi_1(T^4) \cong \mathbb{Z}^4$.
2. The energy of a bit follows from $S = h$ and $\Delta x \Delta p \sim \hbar$: $E_{\text{bit}}(L) = \hbar c/L$.
3. Landauer $E_L = k_B T \ln 2$ is the special case at biological room temperature ($L \sim 10^{-5}$ m, $T \sim 300$ K).
4. All phase transitions are discrete because $\pi_1(T^4) \cong \mathbb{Z}^4$ is integer-valued.
5. Geometry is more fundamental than energy: $T^4 \to \Delta w \to E_{\text{bit}}(L) \to E_L$ (special case).
6. Scale resonance at the scale of life is an open hypothesis, not a derivation.

**Methodological classification:** Dok. 257 is an algebraic bridge from $T^4$ and $\pi_1(T^4) \cong \mathbb{Z}^4$. The scale resonance hypothesis requires further investigation.

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# Bibliography


[1] J. Pascher, *Information as Topologically Frozen Kinetic Energy*, Dok. 255, Zenodo DOI: 10.5281/zenodo.20117635 (2026).

[2] R. Landauer, *Irreversibility and heat generation in the computing process*, IBM Journal of Research and Development **5**(3), 183–191 (1961).

[3] A. Bérut et al., *Experimental verification of Landauer's principle*, Nature **483**, 187–189 (2012).

[4] M. M. Vopson, *The mass-energy-information equivalence principle*, AIP Advances **9**(9), 095206 (2019).